

PHYS 230 - Electricity and Magnetism

Lecture 35

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

9 Electromagnetic Induction

- Lenz's Law (Continued)
- Motional EMF

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

Physics for Scientists and Engineers, Hawkes, et al.

Physics for Scientists and Engineers With Modern Physics, Serway, Jewett

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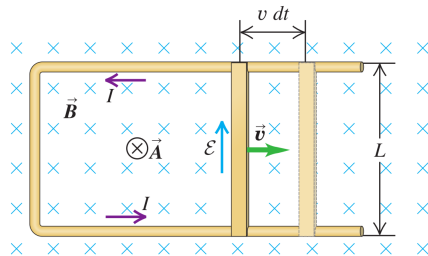
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Exercise 1

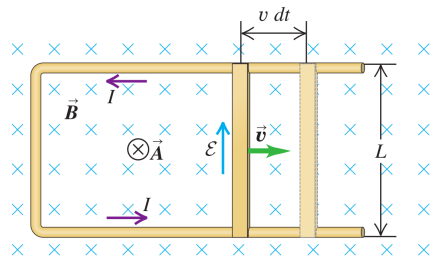
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Figure shows a U-shaped conductor in a uniform magnetic field $\mathbf{B}$ perpendicular to the plane of the figure and directed into the page. We lay a metal rod (the “slidewire”) with length L across the two arms of the conductor, forming a circuit, and move it to the right with constant velocity $\mathbf{v}$. This induces an emf and a current, which is why this device is called a slidewire generator. Find the magnitude and direction of the resulting induced emf.



Exercise I

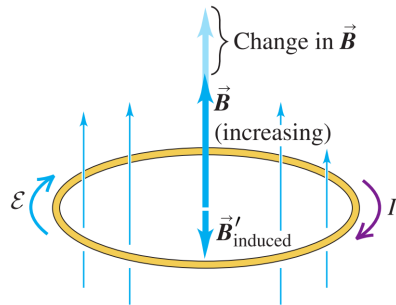
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Exercise 2

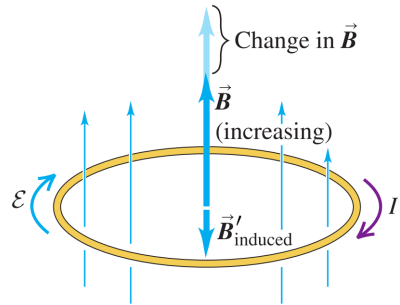
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In figure the magnitude of the magnetic field $\vec{B}$ through the coil is increasing, so there is an induced emf. Use Lenz's law to determine the direction of the resulting induced current.



Exercise 2

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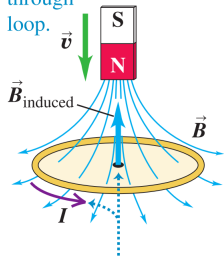


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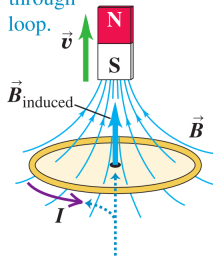
Figure 29.14 Directions of induced currents as a bar magnet moves along the axis of a conducting loop. If the bar magnet is stationary, there is no induced current.

- (a) Motion of magnet causes *increasing downward flux* through loop.

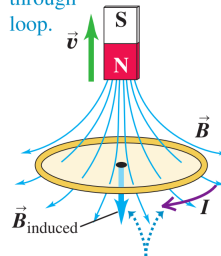


The induced magnetic field is *upward* to oppose the flux change. To produce this induced field, the induced current must be *counterclockwise* as seen from above the loop.

- (b) Motion of magnet causes *decreasing upward flux* through loop.

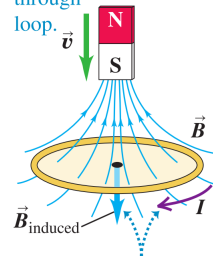


- (c) Motion of magnet causes *decreasing downward flux* through loop.



The induced magnetic field is *downward* to oppose the flux change. To produce this induced field, the induced current must be *clockwise* as seen from above the loop.

- (d) Motion of magnet causes *increasing upward flux* through loop.



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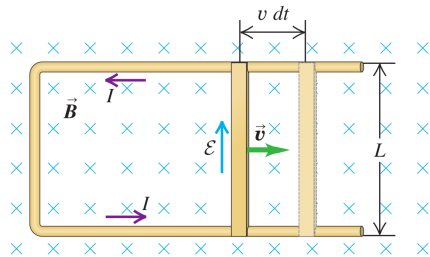
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The Slidewire Revisited

The Faraday's law in vector form in this case can be written as

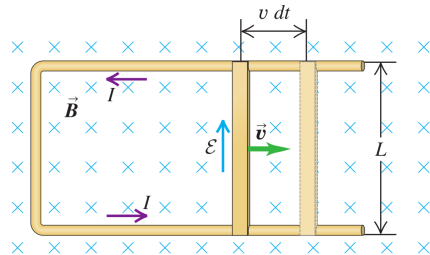
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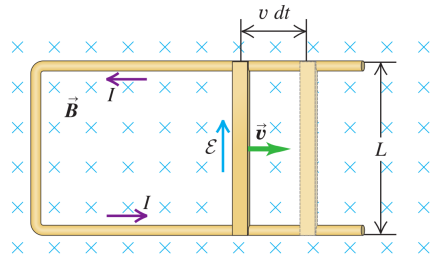
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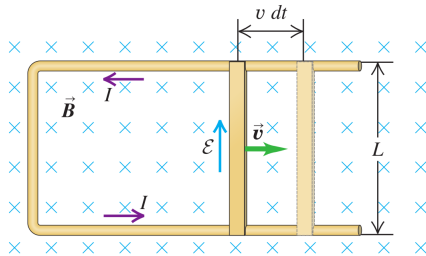
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Let's take $\mathbf{A}$ in the same direction as induced magnetic field $\mathbf{B}'$. Then we can write (w is the width of bottom side of the rectangle and L is the right side)

$$d\mathbf{A} = -d(\mathbf{L} \times \mathbf{w})$$



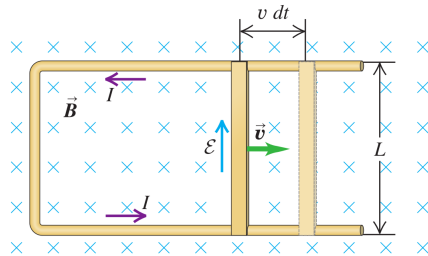
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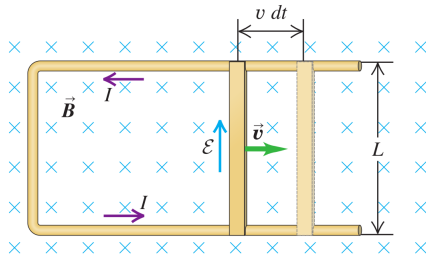
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So we have

$$\mathcal{E} = -\mathbf{B} \cdot \left[-\mathbf{L} \times \underbrace{\frac{d\mathbf{w}}{dt}}_{\mathbf{v}} \right]$$



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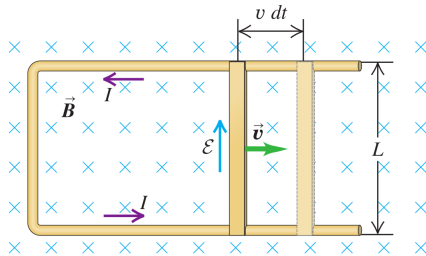
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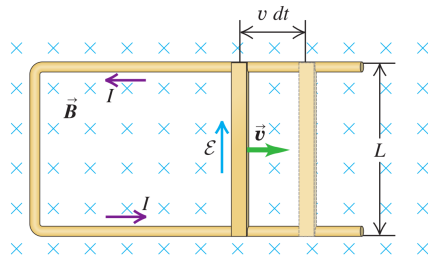
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The Slidewire Revisited

Using the properties of cross product and triple vector product, we can write

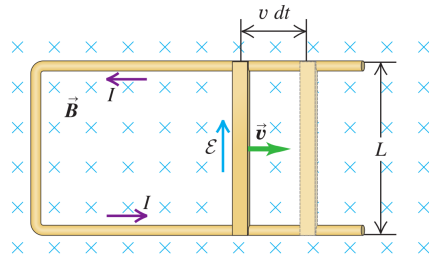
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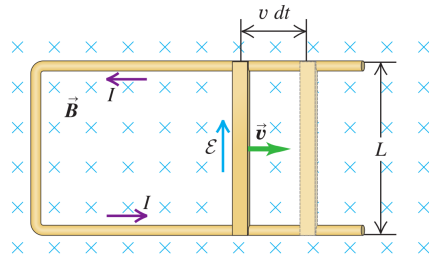
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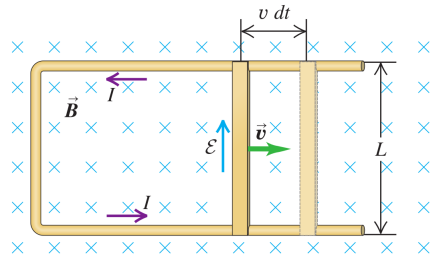
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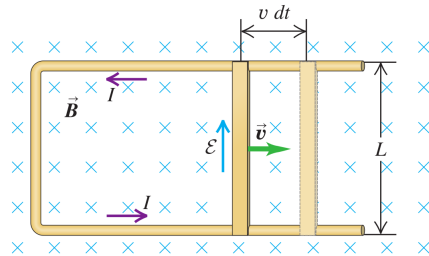
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If we were dealing with an object with infinitesimal $d\mathbf{L}$ we would have obtained

$$d\mathcal{E} = (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{L}$$

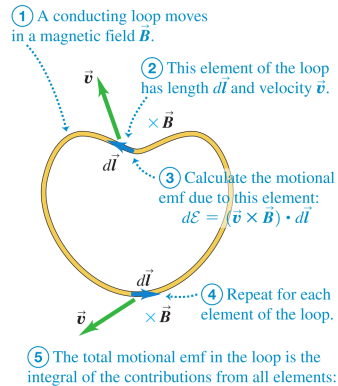


Motional EMF

So for a general moving conductor, in a non-changing $\mathbf{B}$ for each of its infinitesimal elements we have

$$d\mathcal{E} = (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{L}$$

Figure 29.16 Calculating the motional emf for a moving current loop. The velocity $\vec{v}$ can be different for different elements if the loop is rotating or changing shape. The magnetic field $\vec{B}$ can also have different values at different points around the loop.

- 
- The diagram shows a yellow irregularly shaped conducting loop. At two points on the loop, a green vector $\vec{v}$ points upwards and to the right, representing the velocity of the loop. At these same points, a blue vector $d\vec{l}$ is tangent to the loop. A green vector $\vec{v} \times \vec{B}$ is shown pointing upwards and to the left from the first $d\vec{l}$. Dotted blue lines connect numbered callouts to these elements.
- ① A conducting loop moves in a magnetic field $\vec{B}$.
 - ② This element of the loop has length $d\vec{l}$ and velocity $\vec{v}$.
 - ③ Calculate the motional emf due to this element:
 $d\mathcal{E} = (\vec{v} \times \vec{B}) \cdot d\vec{l}$
 - ④ Repeat for each element of the loop.
 - ⑤ The total motional emf in the loop is the integral of the contributions from all elements:

$$\mathcal{E} = \oint (\vec{v} \times \vec{B}) \cdot d\vec{l}$$

Motional EMF

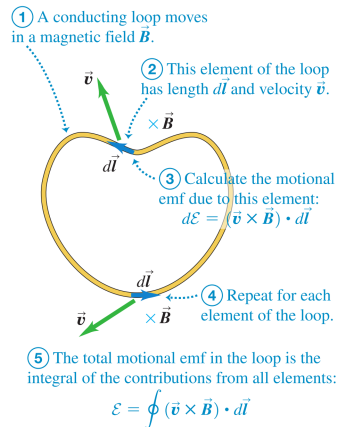
So for a general moving conductor, in a non-changing $\mathbf{B}$ for each of its infinitesimal elements we have

$$d\mathcal{E} = (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{L}$$

and for the whole conductor we integrate over the complete loop of the conductor

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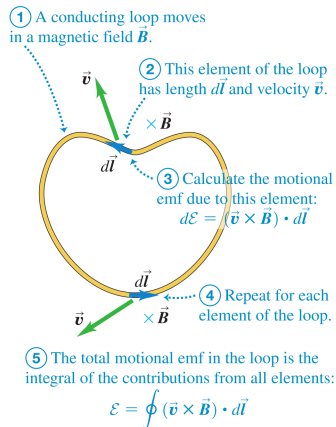
and for the whole conductor we integrate over the complete loop of the conductor

$$\mathcal{E} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{L}$$

If the conductor is stationary, but $\mathbf{B}$ is changing, the above cannot be used;
In that case just use

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

Figure 29.16 Calculating the motional emf for a moving current loop. The velocity $\vec{v}$ can be different for different elements if the loop is rotating or changing shape. The magnetic field $\vec{B}$ can also have different values at different points around the loop.



Direction and sign of $\mathcal{E}$

Use

$$\mathcal{E} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{L}$$

only in case of a moving conductor, in a non-changing $\mathbf{B}$:

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- Instead, we choose an arbitrary direction for induced I and thus $d\mathbf{L}$.

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- If $\mathcal{E}$ turns out to be positive, it means we chose the right direction for I from the beginning.

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- If $\mathcal{E}$ turns out to be positive, it means we chose the right direction for I from the beginning.
- If $\mathcal{E}$ turns out to be negative, it means we chose the wrong direction for I , they are inverse of what we assumed.

Direction and sign of $\mathcal{E}$

Use

$$\mathcal{E} = -\frac{d\Phi_B}{dt}$$

for a conductor that is stationary, in a changing **B**:

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- A positive $\mathcal{E}$ means the induced $\mathbf{B}'$ is in the direction/opposite to the original $\mathbf{A}$ and from there we find the direction of $\mathcal{E}$ and I

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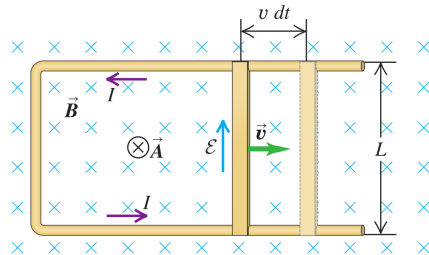
for a conductor that is stationary, in a changing $\mathbf{B}$:

- A positive $\mathcal{E}$ means the induced $\mathbf{B}'$ is in the direction/opposite to the original $\mathbf{A}$ and from there we find the direction of $\mathcal{E}$ and I
- A negative $\mathcal{E}$ means the induced $\mathbf{B}'$ is in the opposite direction to the original $\mathbf{A}$ and from there we find the direction of $\mathcal{E}$ and I

Exercise 3

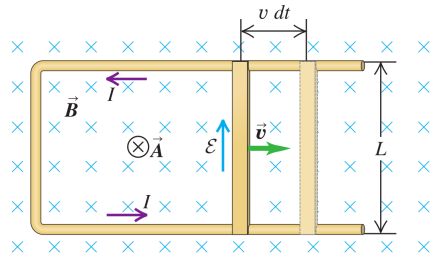
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Suppose the moving rod in figure is 0.10 m long, the velocity v is 2.5 m/s, the total resistance of the loop is $0.03\ \Omega$, and B is 0.60 T. Find the motional EMF, the induced current, and the force acting on the rod.



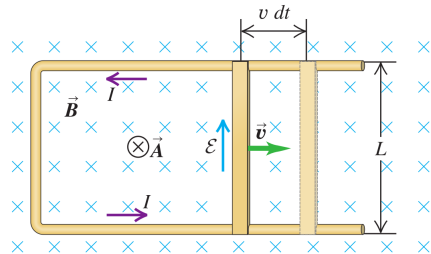
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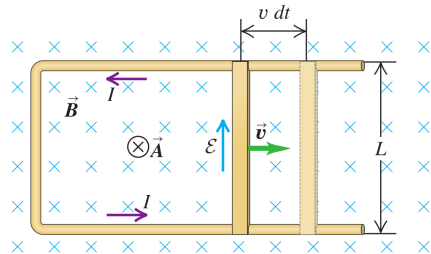
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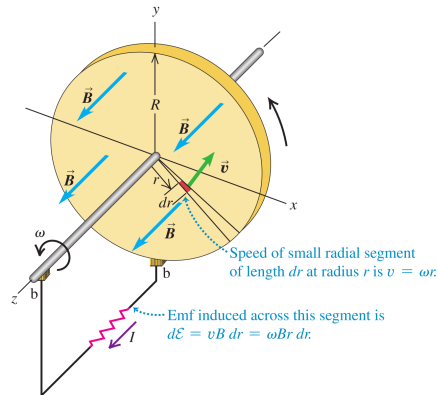


Exercise 4

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Figure shows a conducting disk with radius R that lies in the xy -plane and rotates with constant angular velocity ω about the z -axis. The disk is in a uniform, constant $\mathbf{B}$ field in the z -direction. Find the induced emf between the center and the rim of the disk.

Figure 29.17 A conducting disk with radius R rotating at an angular speed ω in a magnetic field $\mathbf{B}$. The emf is induced along radial lines of the disk and is applied to an external circuit through the two sliding contacts labeled b.



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